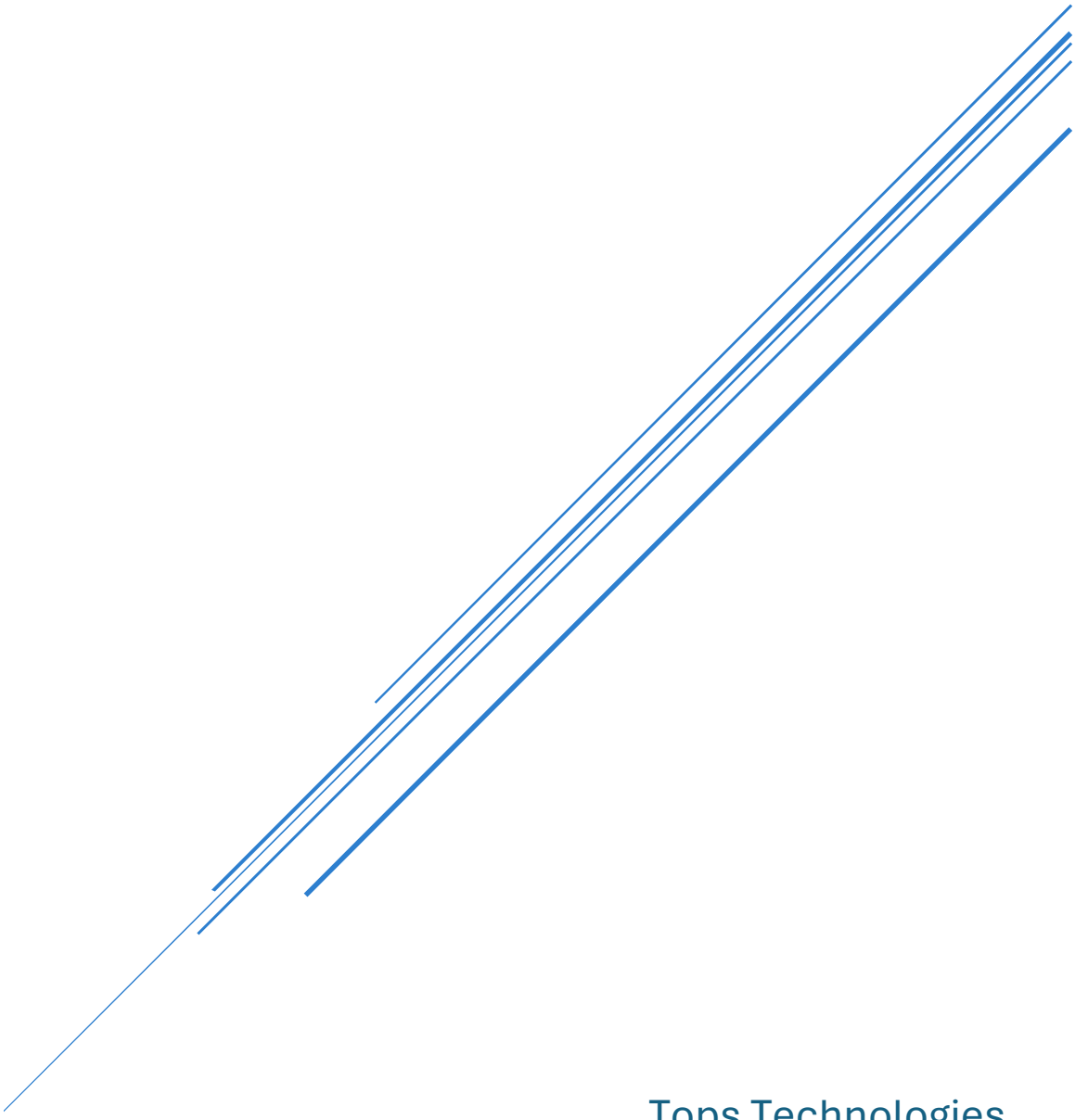
ETHERNET STANDERD AND SPECIFICATION

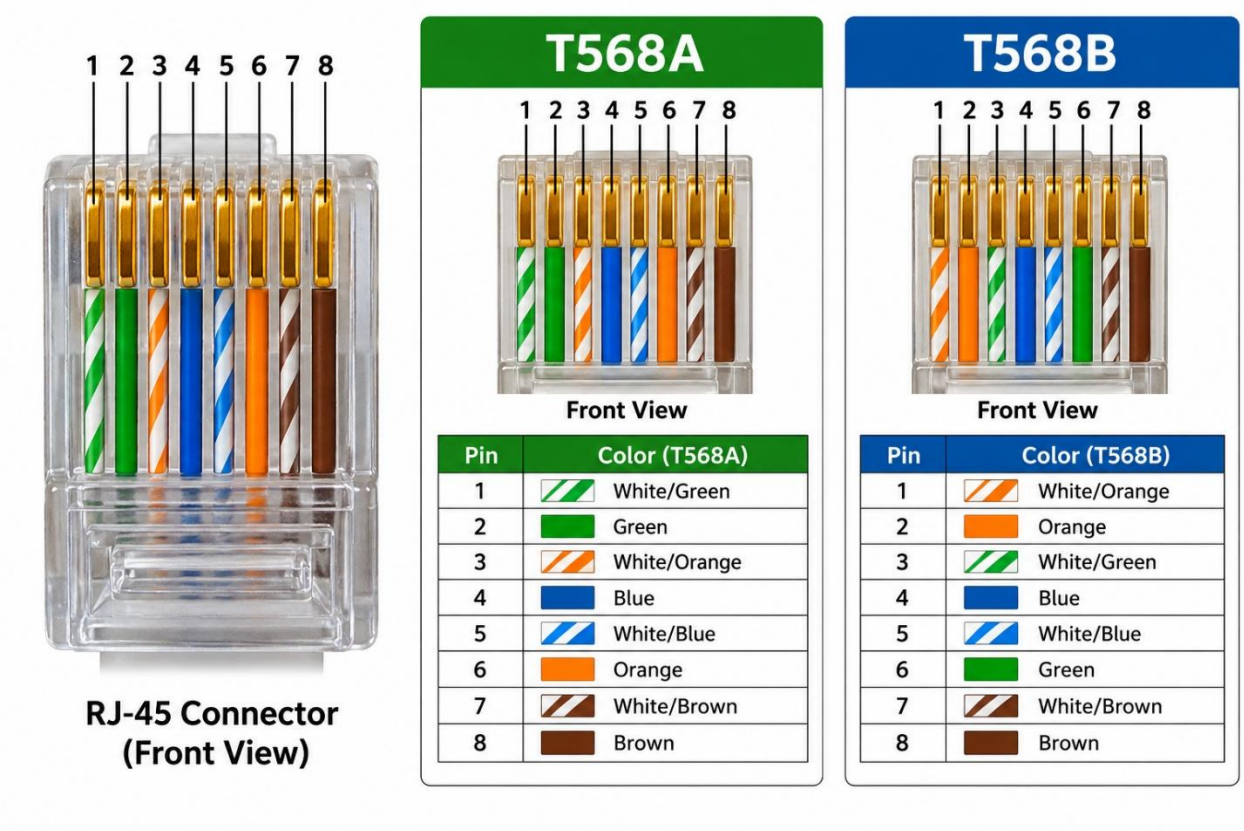
Assignment - 5



Tops Technologies
Cyber Security

1. **Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.**

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2. **Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.**

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➢ Comparison of Ethernet Standards

Specification	10BASE-T	100BASE-TX	1000BASE-T
Speed	10 Mbps	100 Mbps	1000 Mbps (1 Gbps)
Cable type	UTP Cat 3 or higher	UTP Cat 5 or higher	UTP Cat 5e recommended (Cat 5 can support it under specified conditions)
Maximum cable length	100 metres	100 metres	100 metres
Connector type	RJ-45 (8-pin)	RJ-45 (8-pin)	RJ-45 (8-pin)
Ethernet type	Standard Ethernet	Fast Ethernet	Gigabit Ethernet
Pairs used	2 pairs	2 pairs	4 pairs

3. **Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.**
   Hint: Think about how transmit and receive pairs are wired in each cable type.

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To connect two laptops directly using Ethernet, I would use an Ethernet crossover cable.

Reason: In a crossover cable, the transmit (TX) and receive (RX) wire pairs are crossed, so one laptop's transmit pair connects to the other laptop's receive pair. This allows the two laptops to communicate directly.

A straight-through cable has the same wiring standard at both ends and is traditionally used to connect different types of devices, such as a laptop to a switch.

4. **List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.**

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Device	Ethernet Standard	Reason
1. Wi-Fi Router	1000BASE-T	Many modern routers have Gigabit Ethernet ports.
2. Desktop Computer	1000BASE-T	Most modern desktops support Gigabit Ethernet.
3. Network Printer	100BASE-TX	Many network printers support Fast Ethernet; some newer models support Gigabit Ethernet.